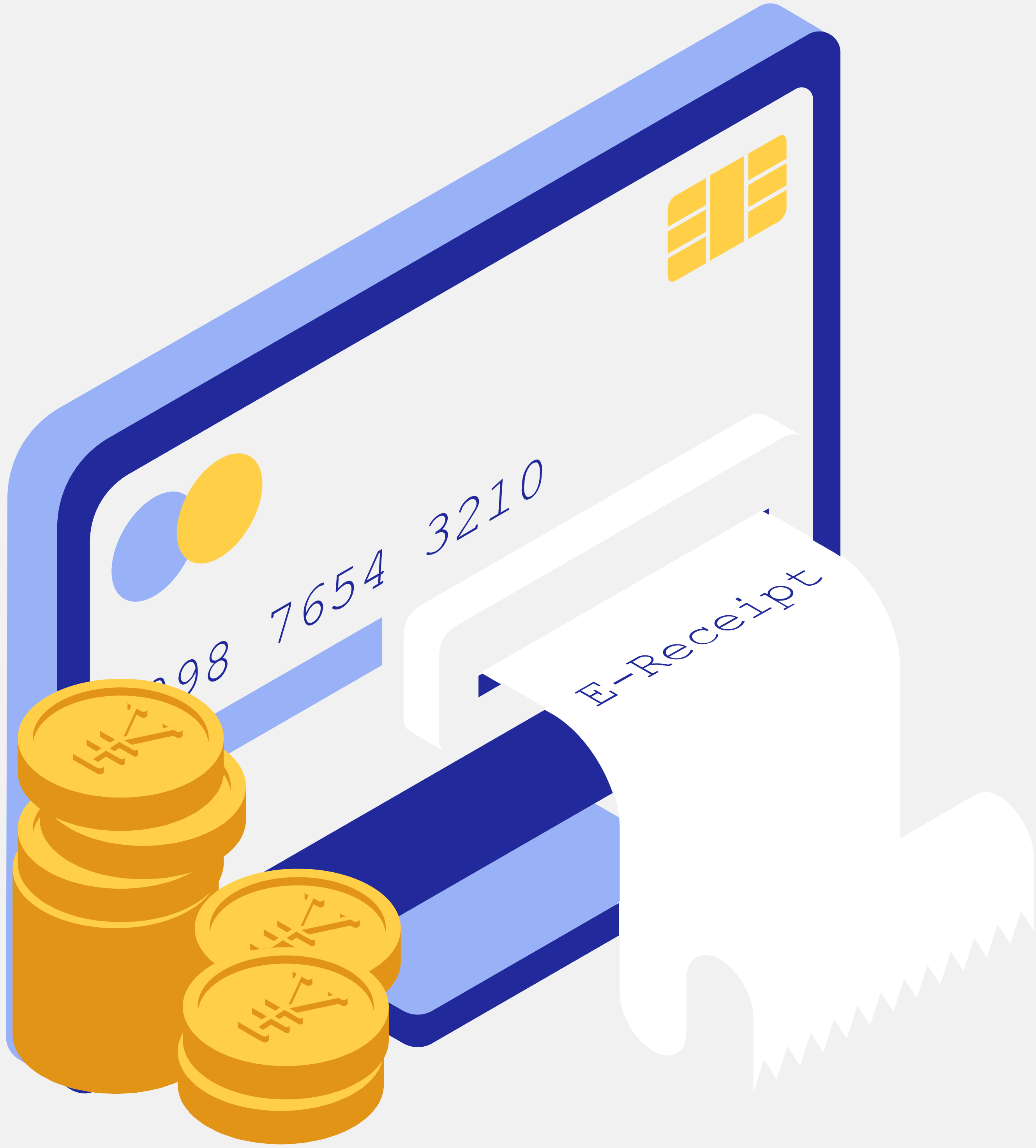

Bank Customer Churn Analysis

Bank Customer Retention Analysis using Microsoft Excel

Prepared by :
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Business Background

ABC Bank has experienced a rise in customer churn, resulting in reduced revenue, increased customer acquisition costs, and lower long-term profitability. Management requires a data-driven understanding of:

- Which customers are leaving
- Why they are leaving
- How churn can be reduced through targeted retention strategies

Objective

Analyze customer demographic and account data to identify the key drivers of customer churn and provide actionable business recommendations that improve customer retention.

Business Questions

1. Which customers are most likely to churn?
2. Which customer characteristics influence churn?
3. Does geography affect churn?
4. Does age influence churn?
5. Does account balance affect churn?
6. Does the number of products impact churn?
7. Are inactive customers more likely to leave?
8. Does owning a credit card reduce churn?
9. Does customer tenure influence churn?
10. What actions can reduce customer churn?

Dataset Overview

Attribute	Details
Domain	Banking
Dataset Source	Kaggle – Bank Customer Churn Dataset
Dataset Size	10,000 Customer Records
Target Variable	Exited (Customer Churn)
Integrated Dataset	Customer_Churn_Master
Analysis Tool	Microsoft Excel

Methodology

Tools Used

- Microsoft Excel
- Power Query
- Pivot Tables
- Pivot Charts
- XLOOKUP
- Slicers

Analysis Techniques

- Data Cleaning & Validation
- Data Transformation
- Data Integration
- Customer Segmentation
- KPI Development
- Interactive Dashboard Design

Data Preparation & Quality Assessment

Data Cleaning

✔️ Removed duplicate records

✔️ Standardized Geography values

✔️ Corrected invalid salary placeholders

✔️ Validated missing Customer IDs

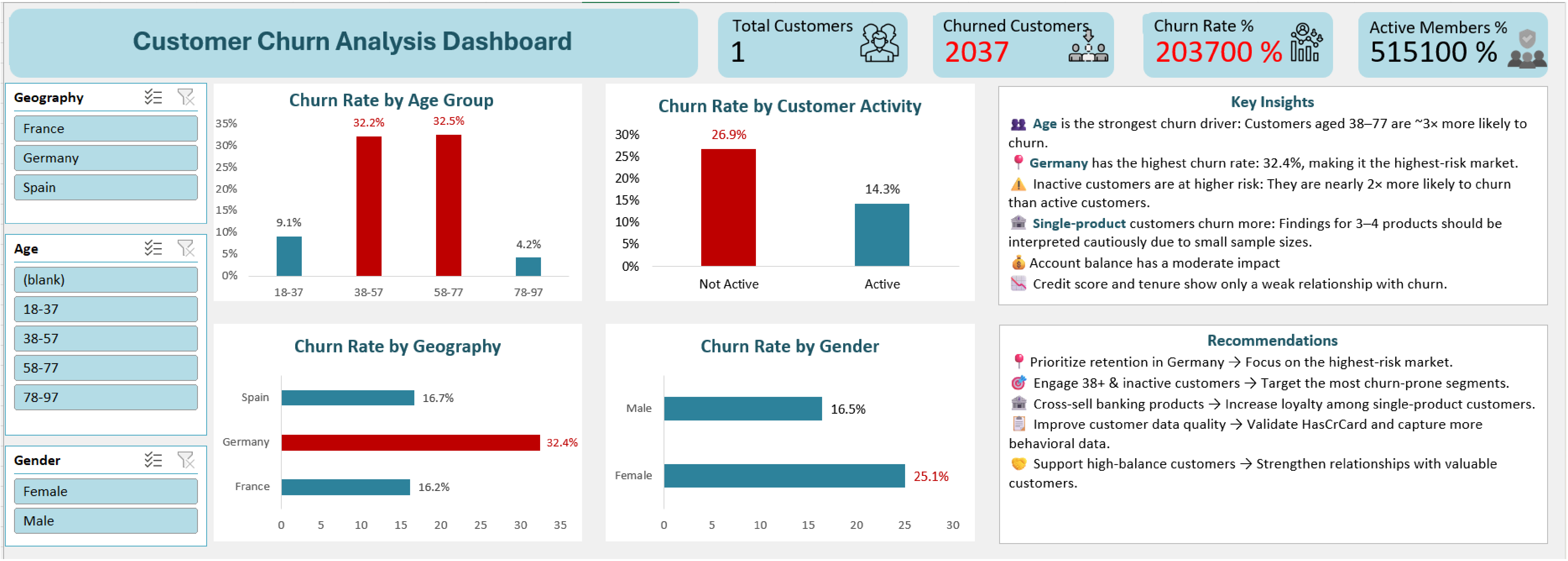
✔️ Preserved non-critical missing surname records

Data Quality Limitation

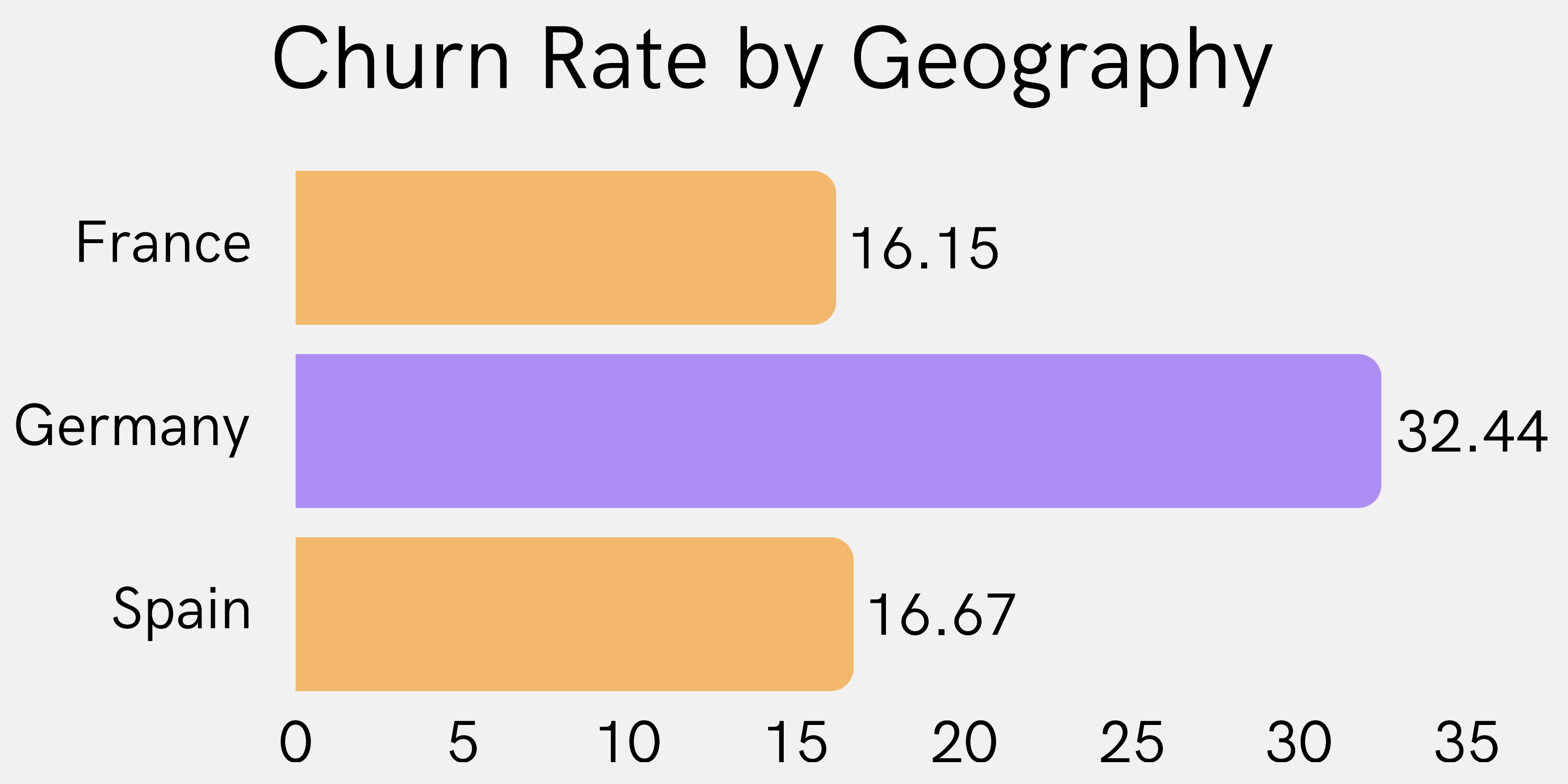
- The HasCrCard and IsActiveMember columns contain identical values throughout the dataset.
- This appears to be a source-system issue and limits independent analysis of credit card ownership.

Dashboard

Interactive dashboard enabling dynamic customer segmentation using Geography, Age Group and Gender filters.

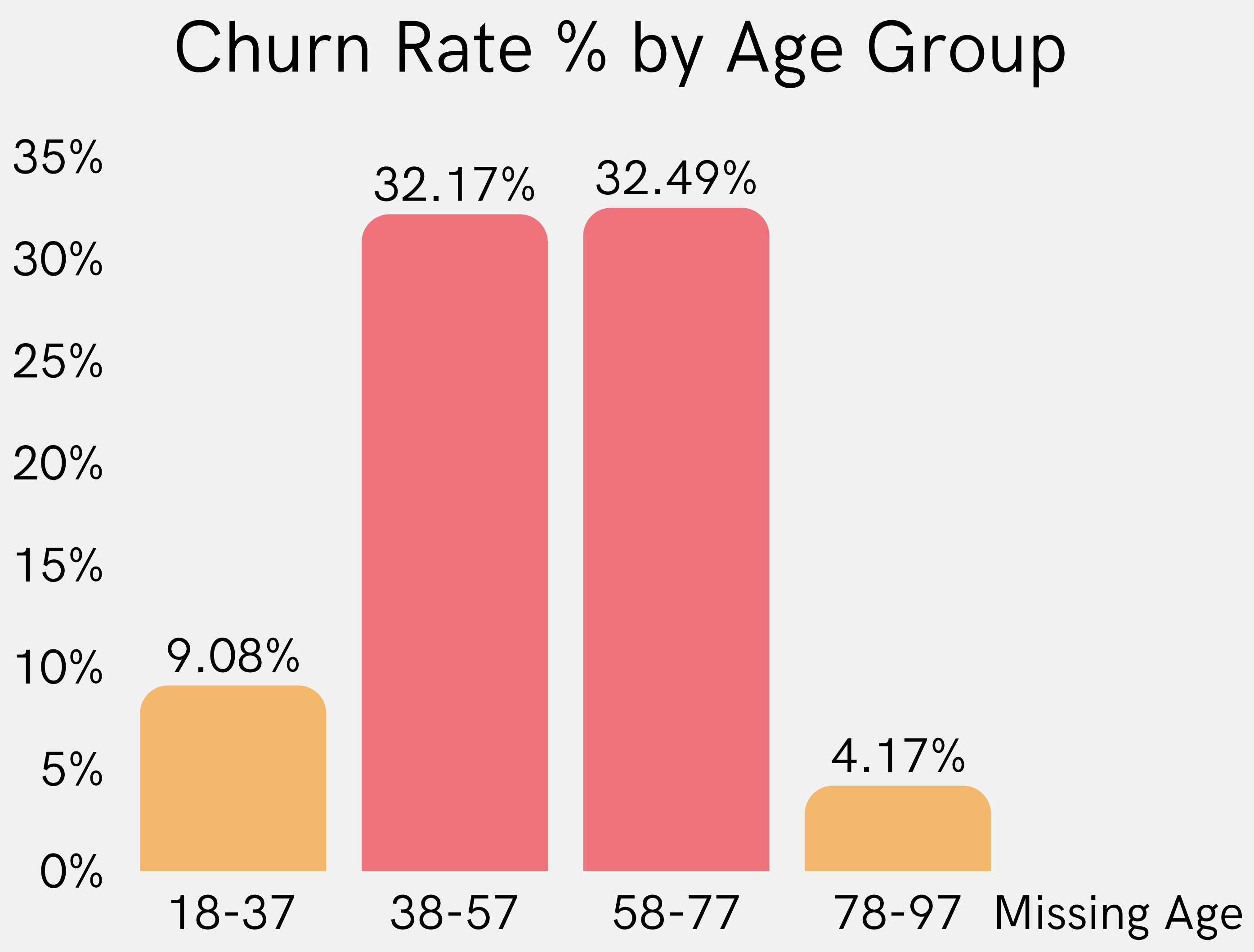


Executive Insights



1. Germany has the highest churn

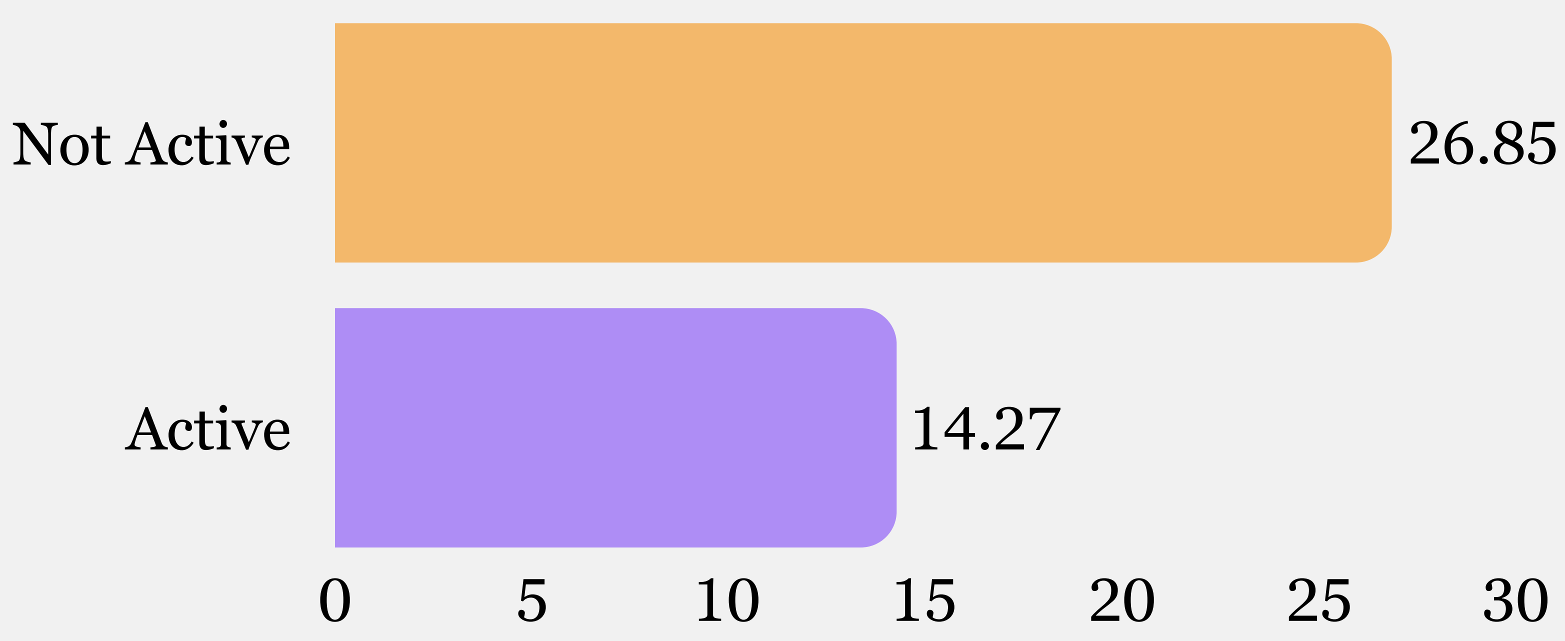
Germany records a churn rate of 32.4%, approximately twice that of France and Spain.



2. Age is the strongest churn driver

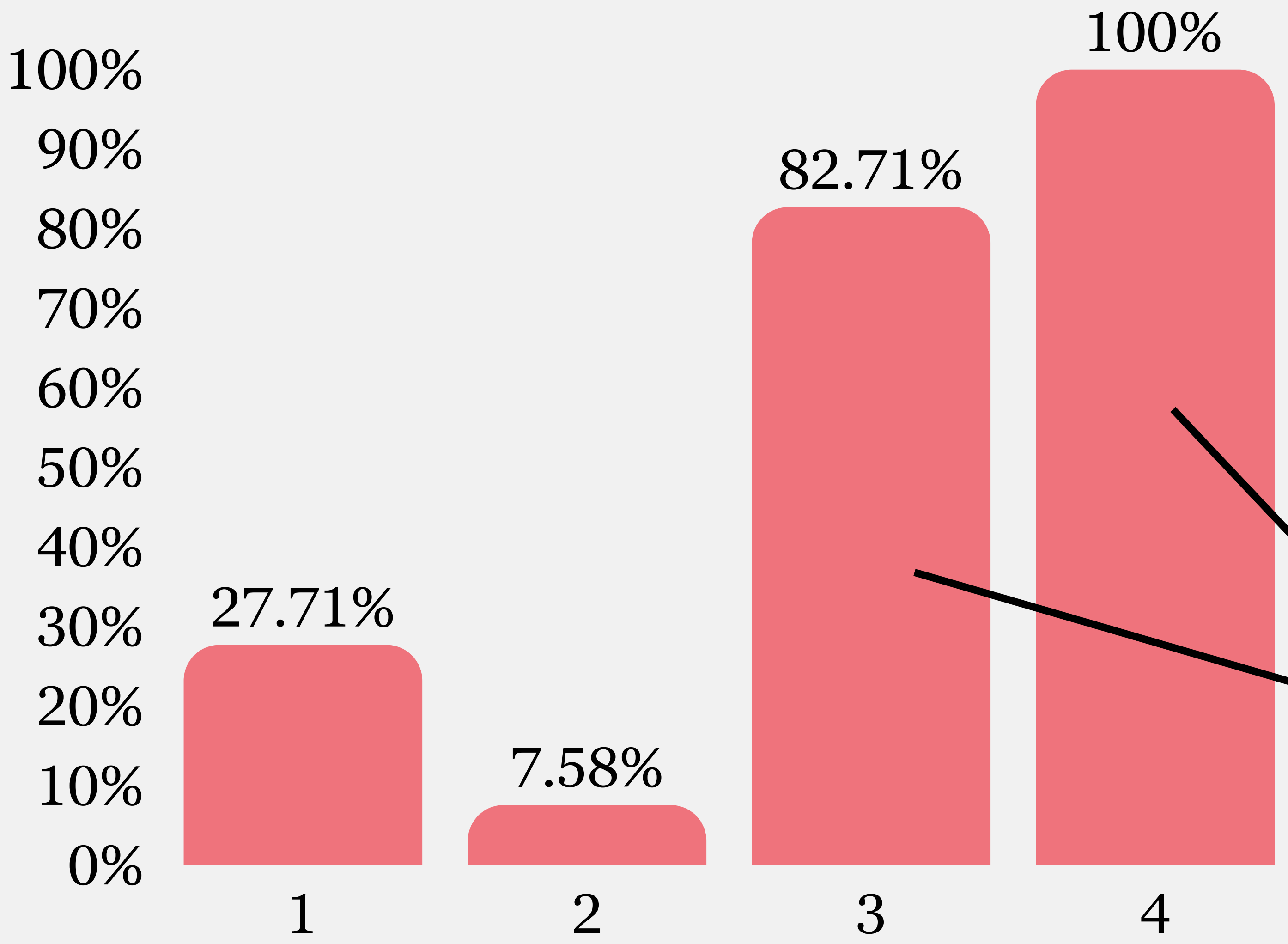
Customers aged 38–77 years exhibit approximately 3× higher churn than younger customers.

Executive Insights



3. Customer inactivity significantly increases churn

Inactive customers are nearly 2× more likely to churn.



4. Single-product customers show higher churn

Customers with one banking product exhibit considerably higher churn than customers with two products.

NOTE :

3–4 no. of products is based on a very small sample size. Additional data and further investigation are required before drawing business conclusions.

Executive Insights

5. Account balance shows a moderate relationship with churn
Customers with balances between €100K–€150K demonstrate moderately higher churn, while higher balance groups require additional data for reliable analysis.

Germany Deeper Analysis

Germany represents the highest-risk market.
Characteristics of high-risk customers include:

- Age above 38 years
- Inactive members
- Single-product customers
- Female customers

This segment should be prioritized for future retention initiatives.

Business Recommendations

Immediate

- Prioritize retention campaigns in Germany.
- Target inactive customers with personalized engagement.
- Focus loyalty programs on customers aged 38+.

Medium-Term

- Promote multi-product adoption through cross-selling.
- Strengthen relationship management for higher-balance customers.

Long-Term

- Improve source-system data quality.
- Capture behavioural metrics (transactions, digital engagement, complaints) to improve churn prediction.

Questions & Discussion

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